

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

Since the Yellow Taxi data for **2023** is stored as **12 separate Parquet files (one per month)**, the first step is to load all these files and combine them into a single DataFrame for analysis.

Approach

1. Use **pandas** with a Parquet engine (pyarrow or fastparquet)
2. Read all monthly Parquet files for 2023
3. Concatenate them into one DataFrame
4. Perform a quick sanity check (shape, columns, sample rows)
5. After merging all 12 parquet files, we got **Rows = 37928560 & columns = 20**
6. And columns are : VendorID', 'tpep_pickup_datetime', 'tpep_dropoff_datetime', 'passenger_count', 'trip_distance', 'RatecodeID', 'store_and_fwd_flag', 'PULocationID', 'DOLocationID', 'payment_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'total_amount', 'congestion_surcharge', 'airport_fee', 'Airport_fee'

1.1.1. Sample the data and combine the files

Approach

1. Convert datetime and extract date & hour using 'tpep_pickup_datetime' column from df dataframe and extract date & time
2. Perform 5% stratified sampling
3. Validate the sampling. Print shape of original dataset and samples dataset
4. This gives us output as **Original dataset size: (37928560, 22) and Sampled dataset size: (1896445, 22)**
5. The pickup timestamp was decomposed into date and hour components. Stratified random sampling was then applied by selecting 5% of trips from each hour of every day. This approach preserves the temporal distribution of taxi demand while reducing data size for efficient exploratory analysis.

2. Data Cleaning

Now we have loaded sampled parquet file to data frame df with rows = 1896445
& columns = 20

2.1. Fixing Columns

2.1.1. Fix the index

Since the dataset was created through concatenation and sampling, resetting the index ensures a clean, continuous index. Useful for Clean row referencing, No confusion during filtering or visualization.

2.1.2. Combine the two airport_fee columns

Inspect missing values in both columns helps to confirm Missing values in one may be present in the other.

APPROACH

- Combine both columns safely. We have used combine_first as Preserves non-null values, avoids overwriting valid data & Handles NaNs correctly
- Drop the redundant column i.e. Airport_fee
- Validate the result

ANALYSIS

- After combining the two airport fee columns, the number of missing values remained unchanged. This indicates that both columns contained null values for the same records. The missing values might correspond to non-airport trips, for which the airport fee is not applicable. Hence, the null values are valid and were retained.
- Null count for Before combining with column Airport Fee 64789
- Null count AFTER combining with column Airport Fee 64789

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

ANALYSIS

The missing value analysis revealed that only five columns contained missing values [RatecodeID, passenger_count, airport_fee, congestion_surcharge, store_and_fwd_flag]. each with a low and identical proportion of 3.42%. This indicates systematic missingness rather than random data loss. Since the missing values are either contextually valid (airport and congestion charges) or low-impact fields, appropriate column-wise imputation and retention strategies were applied.

A uniform missingness pattern across selected columns indicates structured data absence rather than data corruption

2.2.2. Handling missing values in passenger_count

ANALYSIS

The passenger_count column contained missing values and zero entries. Since a taxi trip cannot have zero passengers, such values were treated as missing. All missing values were then imputed using the median passenger count, which represents the most typical trip size and is robust to outliers

Why we imputed using median

- passenger_count is **discrete**
- Median is **robust to outliers**
- Typical taxi trips have **1 passenger**

2.2.3. Handle missing values in RatecodeID

RatecodeID (categorical) had **64,789 missing values**. Missing values in RatecodeID were handled effectively using mode imputation, ensuring data completeness while maintaining business and statistical integrity

2.2.4. Impute NaN in congestion_surcharge

Missing values in the congestion_surcharge column were imputed with zero, as the surcharge is applicable only to certain trips and a missing value indicates that no congestion fee was charged.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

- Check summary statistics of entire data
- Key Outliers Identified are:

Trip distance (These are data errors (GPS, meter, or system glitches))

- **Min = 0**
- **Max = 126,360 miles** ✗ (physically impossible)
- **Mean = 3.89, Median = 1.79**

Fare amount

- **Max = 143,163** ✗
- Typical NYC taxi fares rarely exceed **\$300**

Total amount

- **Max = 143,167** ✗
- Closely tied to fare → same anomaly source

Tolls amount

- **Max = 143** ✗
- Realistic tolls usually < **\$50**

Tip amount

- **Max = 223** ✗
- Tips above fare itself are suspicious

RatecodeID

- **Max = 99** ✗
- Valid values are **1–6 only**

Passenger count

- **Max = 9** ✗
- NYC yellow taxis allow **1–6**

Airport fee

- Mean $\approx$ **0.14**
- Valid values are typically **0 or 1.25**
- Std > mean $\rightarrow$ inconsistent entries

Columns That Look Fine (No Action Needed)

- ✓ VendorID (1–2)
- ✓ MTA tax (mostly 0.5)
- ✓ Improvement surcharge (mostly 1.0)
- ✓ Congestion surcharge (mostly 2.5)
- ✓ PU / DO Location IDs

Strategy to Manage Outliers (Best Practice)

We **DO NOT delete blindly**. We use **domain rules + capping**.

Step 1: Fix logical errors (hard filters)

Step 2: Cap monetary outliers

Step 3: Airport fee correction

Step 4: Recheck consistency of total amount

ANALYSIS

Outliers were handled using **domain-driven rules** rather than purely statistical methods. Impossible values (e.g., trip distance >100 miles, RatecodeID outside 1–6) were removed, while extreme but plausible monetary values were capped to preserve data volume and revenue integrity.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Numerical :

passenger_count, trip_distance, fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, congestion_surcharge, airport_fee, total_amount

Categorical

- ☐ VendorID (*ID – categorical despite int64*)
- ☐ RatecodeID (*fare type – categorical*)
- ☐ store_and_fwd_flag (*Y/N*)
- ☐ payment_type (*payment mode – categorical*)
- ☐ PULocationID (*zone ID – categorical*)
- ☐ DOLocationID (*zone ID – categorical*)

Datetime

tpep_pickup_datetime

tpep_dropoff_datetime

- 3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

Taxi Pick up by Hours

Taxi pickups show a highly predictable hourly distribution, with minimal demand in early mornings, steady growth during the day, and a strong evening peak. This makes hourly segmentation critical for operational planning, pricing, and resource optimization

Taxi Pick up by Days of the week

Taxi pickups show a strong weekday-driven demand, peaking on Thursday, with a moderate decline toward the weekend. Sundays consistently exhibit lower demand, highlighting the importance of day-of-week-based planning for efficient taxi operations

Taxi Pick up by Months

Monthly taxi pickups exhibit strong **seasonality**, with highest demand in **spring and autumn**, and noticeably lower demand in **mid-summer**. Understanding these patterns helps optimize **fleet management, pricing, and resource planning**

3.1.3. Filter out the zero/negative values in fares, distance and tips

The dataset is free of negative values, indicating good base quality. However, zero values require selective treatment—valid for tips, questionable for distance, and invalid for fare and total amount. Careful filtering will improve both business insight accuracy and model performance.

3.1.4. Analyse the monthly revenue trends

Monthly taxi revenue exhibits strong seasonality, with distinct spring and autumn peaks and mid-year slowdown. This pattern highlights the importance of season-aware pricing, capacity planning, and operational strategy.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue Taxi revenue is seasonally driven, with Q4 as the strongest quarter and Q3 as the weakest. Understanding quarter-wise contribution enables better forecasting, staffing, and pricing strategies.

3.1.6. Analyse and visualise the relationship between distance and fare amount

Correlation is 0.0054 (Almost ZERO). This implies no linear relationship between them. This is unrealistic for taxi data. In real world scenario, correlation between distance and fare amount should be positive. Root cause why correlation is zero:

- Presence of extreme distance and fare outliers
- Possibly unit issues

This tells us that >

- Current correlation is not trustworthy
- Scatter plot is dominated by bad data
- Ideally distance should drive fare which is not the case here
- Data cleaning required

- Cap trip distance & fare amount and then recalculate correlation
- Replot
- Updated correlation post data cleaning is 0.94 which now shows positive linear relationship between trip distance and fare amount

ANALYSIS

Initial analysis showed near-zero correlation due to extreme outliers. After identifying unrealistic trip distances and fares, data cleaning was performed. Post-cleaning, a strong positive relationship emerged between trip distance and fare, aligning with expected taxi fare mechanics

3.1.7. Analyse the relationship between fare/tips and trips/passengers

Relationship between fare/tips

Trip duration shows a strong positive correlation (0.81) with fare amount, indicating that time-based factors such as congestion significantly influence pricing. Longer trips—especially during traffic—generate higher fares even when distance is limited.

relationship between tips/ passengers

Correlation is 0.041 indicates Passenger count has no meaningful linear relationship with fare amount

Boxplot analysis :

- Median fare across 1-6 passengers is nearly constant
- Strong evidence that pricing does not depend on passenger count
- Spread IQR is almost similar across all passenger groups
- Indicates fare variability is driven by distance, duration & surcharge
- Outliers exists for all groups
- High-fare outliers appear for every passenger count
- These corresponds to long distance & long duration trips, airport or toll heavy trips and not because of passenger counts
- Fewer trips for 5-6 passenger counts
- Passenger count does not influence taxi fare; distance, duration, and congestion are the true revenue drivers

Relationship between tip / trip distance

- Correlation between tip amount and trip distance is 0.798 indicated strong positive relationship
- As trip distance increases, tip amount increases as well, however this is not a pure distance effect.
- Scatter plot shows clear upward trend — but with heavy dispersion
- This indicates Longer trips tend to have higher tips, But for the *same distance*, tips vary widely, Indicates human behavior, not rule-based pricing
- Dense cluster at short distances shows Majority of trips are under 20 miles, Tip amounts mostly below 50, Long-distance trips are fewer but show larger tips
- Presence of high-tip outliers: Tips exceeding 100–200, Likely Business travelers, Airport / long-haul trips, Card-based tipping with suggested percentages

ANALYSIS: Tip amount shows a strong positive correlation with trip distance (0.80). This relationship is largely indirect, as longer trips generate higher fares, which in turn lead to higher tips. Tips increase with trip distance, primarily because longer trips cost more—not because distance itself influences tipping behavior.

3.1.8. Analyse the distribution of different payment types

- Credit Card Dominates. Clear majority of trips are **cashless**
- Cash Is Still Significant (~17%). One in **every 6 trips** is paid in cash
- Cash trips are Often short distance or Often zero recorded tips (even if tipped)
- This introduces **tip bias** in the dataset
- Payment Type = 0 (3.42%) → Data Quality Issue. “0” is **not a valid payment type. Likely represents: Missing or Improperly recorded values. Should be excluded from financial & tip analysis**
- Disputes & No-Charge Trips Are Negligible. Combined < **1.3%**
- Usually: Refunds, System corrections or free rides
- Should not be used for: Revenue modeling, Pricing analysis

ANALYSIS

Credit card payments dominate the dataset (78.7%), followed by cash (16.7%). Other payment types form a negligible share and were excluded from downstream financial analysis. Most taxi trips are cashless, enabling reliable revenue and tipping insights, while cash payments still

represent a meaningful secondary mode. Payment behavior is heavily skewed toward digital payments, which significantly improves the reliability of fare and tip analytics.

3.1.9. Load the taxi zones shapefile and display it

NYC taxi zones shapefile was validated visually and structurally. All 263 zones were present with complete geometry, enabling spatial joins with pickup and dropoff location IDs for geospatial demand analysis

3.1.10. Merge the zone data with trips data

Summary of Missing Values After Zone Merge

- Trip-Level Missing Values (Pre-existing in Trip Data). These rows were already incomplete before the spatial join. Impute or drop depending on analysis. For EDA: safe to drop or flag

Column	Missing Count	Interpretation
passenger_count	64,802	Missing passenger info
RatecodeID	64,802	Often tied to same rows
store_and_fwd_flag	64,802	Same rows → systemic
congestion_surcharge	64,802	Not applicable for older trips

- Airport Fee Columns (Schema / Version Issue). Correction is required

Column	Missing
airport_fee	1,748,084
Airport_fee	213,305

- Zone file missing values

Column	Missing
OBJECTID	19,086
Shape_Leng	19,086
Shape_Area	19,086
zone	19,086
borough	19,086
geometry	19,086

-
- Drop these rows with missing values

3.1.11. Find the number of trips for each zone/location ID

- Overall Pattern : Manhattan dominates the top pickup zones, Airports (JFK & LaGuardia) are the only non-Manhattan zones in the Top 10. This indicates two strong demand drivers: Daily urban mobility (office, leisure, tourism) and Airport-related travel
- Top Pickup Zones – Key Insights: Airports as major demand hubs, JFK Airport (Queens) – 97,035 trips (Rank #1), LaGuardia Airport (Queens) – 64,572 trips
- Airports generate high-volume, consistent demand
- Trips from airports usually have : Longer distances, Higher fares and Higher tips
- Manhattan Core : Top Manhattan zones include: Upper East Side (South & North), Midtown Center, Midtown East
- Operational & Business Implications : Prioritize driver supply in Midtown, Upper East Side, Penn Station area and Airports
- It will Reduces passenger wait time and Improves driver earnings consistency

- **Revenue Optimization: Airports** → focus on: Premium services and Upselling

ANALYSIS: Taxi demand is heavily concentrated in Manhattan's commercial and tourist zones, with airports acting as high-revenue outliers. Any operational, pricing, or forecasting strategy must prioritize Midtown Manhattan and airport corridors to maximize efficiency and revenue.

3.1.12. Add the number of trips for each zone to the zones dataframe

Conceptual Steps:

- We already have trip_with_zones → trip-level data + zone attributes + geometry
- Grouped data → number of trips per LocationID / zone
- We will:
 1. Aggregate trips at zone level
 2. Create a zone-level GeoDataFrame (one row per zone)
 3. Merge trip counts into the GeoDataFrame
 4. Use this to plot a choropleth map of total trips per zone
- Now, Group trips by pickup location (zone level). This gives total pickups per taxi zone
- Create a zone GeoDataFrame (one polygon per zone). Without drop_duplicates, the same polygon would appear thousands of times. This step converts trip data → spatial data
- Merge trip counts into the GeoDataFrame
- Plot map – Total Trips per Zone

3.1.13. Plot a map of the zones showing number of trips

- High-intensity zones (dark shades) : Midtown Manhattan, Upper East / West Side , JFK & LaGuardia Airports
- Low-intensity zones (light shades): Peripheral boroughs, Residential / low-density areas
- **Demand concentration** is extremely skewed toward Manhattan
- **Airports act as demand anchors**, even with fewer zones

3.1.14. Conclude with results

- Below are clean, safe ways to display the zones DataFrame sorted by number of trips, both for inspection and reporting.
- Sort zones by number of trips (descending)
- Display key columns only (recommended)
- Display bottom zones
- Reset index for clean presentation
- Plot shows that taxi demand is overwhelmingly Manhattan-centric, with airports acting as secondary hubs.
- Airports are the single biggest pickup zones
- Midtown dominates Manhattan demand
- Upper East & Upper West Side are strong secondary hubs
- Sharp drop after top zones
- Keep maximum driver availability in: Midtown Manhattan, Upper East Side and Airports
- Revenue optimization: Airports and Midtown zones likely: Higher fare per trip, Higher tip amounts
- Taxi pickup demand in NYC is extremely concentrated, with Midtown Manhattan and major airports accounting for the majority of high-volume zones. This highlights strong spatial inequality in travel demand and underscores the importance of zone-aware operational and pricing strategies.
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3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

- Define time-of-day buckets
- Aggregate route-level speed by time of day
- Calculate average speed
- Filter unreliable routes
- Find slowest routes per time of day
- Final output

3.2.2. Calculate the hourly number of trips and identify the busy hours

Taxi demand shows a strong and predictable daily pattern, peaking sharply at 6 PM with ~140K trips.

Demand remains high from noon to evening, indicating prime revenue hours, while early morning sees minimal activity.

These patterns support targeted driver incentives, dynamic pricing, and time-aware ETA models to maximize revenue and service quality.”

3.2.3. Scale up the number of trips from above to find the actual number of trips

After scaling for sampling, the busiest five hours collectively account for several million trips, with peak demand at 6 PM alone exceeding one million trips—highlighting the critical importance of peak-hour capacity planning.

3.2.4. Compare hourly traffic on weekdays and weekends

Weekday taxi demand is highly structured and commute-driven with sharp peaks, while weekend demand is leisure-driven,

flatter, and extends late into the night—requiring fundamentally different driver allocation, pricing, and ETA strategies.”

3.2.5. Identify the top 10 zones with high hourly pickups and drops

TOP 10 pick up zones

	zone	pickup_count
119	JFK Airport	97035
228	Upper East Side South	88536
152	Midtown Center	87557
227	Upper East Side North	79790
153	Midtown East	66759
129	LaGuardia Airport	64572
177	Penn Station/Madison Sq West	64297
221	Times Sq/Theatre District	62722
133	Lincoln Square East	62606
161	Murray Hill	55974

TOP 10 DROP OFF ZONES

	zone	dropoff_count
119	JFK Airport	97035
228	Upper East Side South	88536
152	Midtown Center	87557
227	Upper East Side North	79790
153	Midtown East	66759
129	LaGuardia Airport	64572
177	Penn Station/Madison Sq West	64297
221	Times Sq/Theatre District	62722
133	Lincoln Square East	62606
161	Murray Hill	55974

3.2.6. Find the ratio of pickups and dropoffs in each zone

Zones with high pickup/drop-off ratios represent major transit and commercial hubs where taxi demand is outbound-heavy,

while low-ratio zones are primarily residential or restricted areas acting as trip termination points, highlighting strong

directional mobility patterns and the need for strategic fleet rebalancing.

3.2.7. Identify the top zones with high traffic during night hours

Night-time taxi demand in NYC is highly concentrated around nightlife districts, transit hubs, and airports,

with pickups dominated by entertainment areas and drop-offs skewed toward nearby residential neighborhoods.

The strong spatial overlap indicates short, localized trips driven by safety, convenience, and limited public transport availability

3.2.8. Find the revenue share for nighttime and daytime hours

Although daytime hours contribute nearly 88% of total taxi revenue due to high trip volumes, nighttime hours generate only about 12% despite spanning almost one-third of the day.

This highlights an opportunity to optimize nighttime pricing, as demand during these hours is more inelastic and operationally costly

3.2.9. For the different passenger counts, find the average fare per mile per passenger

The fare per mile per passenger decreases sharply as passenger count increases, demonstrating strong economies of scale in taxi pricing. Single-passenger trips are the least cost-efficient,

while shared trips with 3–6 passengers offer the best value. Extremely high values for passenger counts above 7

indicate data anomalies and should be excluded from pricing decisions

3.2.10. Find the average fare per mile by hours of the day and by days of the week

Average fare per mile varies significantly across days of the week, with Sunday and Tuesday exhibiting the highest

fare efficiency due to longer trips and lower congestion, while Saturday and Friday show lower fare per mile driven by short,

localized leisure travel and traffic congestion.

3.2.11. Analyse the average fare per mile for the different vendors

Fare per mile varies significantly by vendor and hour. Vendor 2 consistently earns higher fares,

especially during late-night and early morning hours, reflecting high-value trips such as airport and long-distance rides.

Vendor 1 maintains stable fares suitable for high-volume daytime trips. Vendor 6 shows extreme outliers that should be cleaned.

These patterns suggest a vendor- and time-specific pricing strategy to optimize revenue while managing operational efficiency.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

Fare per mile decreases with increasing trip distance for all vendors. Short trips (0–2 miles) are the most expensive per mile due

to congestion and minimum fares, with Vendor 2 consistently commanding a premium and Vendor 6 showing extreme outliers.

Medium trips (2–5 miles) provide moderate fare efficiency, while long trips (>5 miles) benefit from economies of scale,

reducing fare per mile. These results support a tiered pricing strategy and a vendor-specific operational allocation to maximize

revenue and efficiency

3.2.13. Analyse the tip percentages

Tip behavior varies significantly by time, distance, and passenger composition. Late-night, short-distance, solo-passenger trips generate the highest tip percentages, while long-distance and large-group trips show consistently lower tipping behavior.

tip_category	
High Tip (>25%)	14.59%
Low Tip (<10%)	13.89%
Medium Tip	14.13%

3.2.14. Analyse the trends in passenger count

	day_of_week	avg_passenger_count
1	Monday	1.366645
5	Tuesday	1.338738
6	Wednesday	1.335052
4	Thursday	1.348395
0	Friday	1.406851
2	Saturday	1.488025
3	Sunday	1.472264

3.2.15. Analyse the variation of passenger counts across zones

Top 10 zones by avg passengers				PULocationID	zone	avg_passenger_count
103	111	Green-Wood Cemetery	2.000000			
5	6	Arrochar/Fort Wadsworth	1.875000			
186	195	Red Hook	1.818653			
195	204	Rossville/Woodrow	1.800000			
11	12	Battery Park	1.763387			
55	58	Country Club	1.666667			
169	178	Ocean Parkway South	1.615385			
0	1	Newark Airport	1.603774			
63	66	DUMBO/Vinegar Hill	1.583691			
251	261	World Trade Center	1.551516			
Bottom 10 zones by avg passengers				PULocationID	zone	avg_passenger_count
164	172	New Dorp/Midland Beach	1.0			
148	156	Mariners Harbor	1.0			
175	184	Pelham Bay Park	1.0			
178	187	Port Richmond	1.0			
201	210	Sheepshead Bay	1.0			
205	214	South Beach/Dongan Hills	1.0			
212	221	Stapleton	1.0			
197	206	Saint George/New Brighton	1.0			
242	252	Whitestone	1.0			
241	251	Westerleigh	1.0			

Zones with higher average passenger counts are typically residential, leisure, or airport-linked areas, indicating group-oriented travel. In contrast, several suburban and coastal zones exhibit exclusively solo travel behavior, reflecting commuter-dominated demand. This distinction has direct implications for vehicle deployment, ride-sharing strategies, and pricing optimization.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

Surcharges are applied with varying frequency in NYC taxi trips. The improvement surcharge and MTA tax are nearly universal,

while congestion surcharges affect a large majority of trips. Moderate-frequency surcharges like ‘extra’ apply to about 60% of trips.

Low-frequency surcharges, such as tolls (8%) and airport fees (0.68%), are infrequent but high-margin. Understanding surcharge frequency

helps in revenue forecasting, fleet allocation, and pricing strategy optimization

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

NYC taxi operations should shift from uniform dispatching to a time-zone-route aware strategy.

Focusing supply where and when demand is concentrated—especially Midtown Manhattan and airports during peak hours—will **maximize utilization, reduce congestion losses, and significantly improve driver and passenger outcomes.**

Final Strategic Summary

Dimension	Optimization Focus
Time	Hour-specific fleet sizing
Space	Zone-centric dispatching
Routes	Congestion-aware routing
Flow	Pickup–dropoff rebalancing
Revenue	Airport & peak-hour prioritization
Efficiency	Eliminate low-yield routes

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Strategic cab positioning must be dynamic, zone-aware, and time-sensitive. Concentrating supply in high-demand zones during peak windows, while actively rebalancing vehicles from low-demand areas, can significantly improve utilization and service efficiency.

Strategic Summary

Dimension	Positioning Approach
Time	Hour-based zone prioritization
Space	Tier-based zone allocation
Day Type	Separate weekday/weekend strategies
Passenger Mix	Vehicle-zone matching
Seasonality	Month-aware scaling
Flow	Post-trip rebalancing

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors

Strategic Summary Table

Dimension	Pricing Approach
Time	Peak-hour controlled surges
Route	Congestion-based surcharges
Zone	Demand-tier pricing
Distance	Flat + transparent pricing
Passenger Mix	Group-friendly pricing
Airports	Fixed, predictable premiums
Day Type	Weekday/weekend differentiation